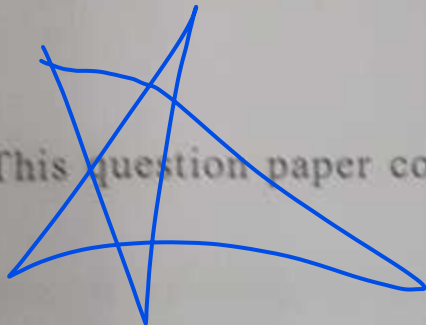


[This question paper contains 8 printed pages.]



Your Roll No.....

Sr. No. of Question Paper : 1409

C

Unique Paper Code : 32351302

Name of the Paper : BMATH306 – Group Theory-I

Name of the Course : B.Sc. (Hons) Mathematics

Semester : III

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. All questions are compulsory.
3. Attempt any two parts from each question from Q2 to Q6.
4. In the question paper, given notations have their usual meaning unless until stated otherwise.

1. Give short answers to the following questions. Attempt any six.

- (i) What is the total no of rotations and total no of reflections in the dihedral group D_3 ? Describe them (rotations and reflections) in pictures or words. What can you say about the total no of rotations and total no of reflections in the dihedral group D_n ?
- (ii) Give one non-trivial, proper subgroup of $GL(2, \mathbb{R})$. Is $GL(2, \mathbb{R})$ a group under addition of matrices? Answer in few lines.
- (iii) Let G be a group with the property that for any a, b, c in G ,
 $ab = ca$ implies $b = c$. Prove that G is Abelian.
- (iv) Give an example of a cyclic group of order 5. Show that a group of order 5 is cyclic.

- (v) Prove that a cyclic group is Abelian. Is the converse true?
- (vi) Find all subgroups of Z_{15} .
- (vii) Prove that 1 and -1 are the only two generators of $(Z, +)$. Give short answer in few lines.
- (viii) " Z_n , $n \in N$, is always cyclic whereas $U(n)$, $n \in N$; $n \geq 2$ may or may not be cyclic". Prove or disprove the statement in a few lines. (6×2=12)
2. (a) Let $G = \{a + b\sqrt{2} \mid a \text{ and } b \text{ are rational nos not both zero}\}$
- Prove that G is a group under ordinary multiplication. Is it Abelian or Non-Abelian? Justify your answer.

(b) Prove that a group of composite order has a non-trivial, proper subgroup.

(c) Prove that order of a cyclic group is equal to the order of its generator. $(2 \times 6.5 = 13)$

3. (a) Prove that every permutation of a finite set can be written as a cycle or as a product of distinct cycles. (6)

(b) (i) In S_4 , write a cyclic subgroup of order 4 and a non-cyclic subgroup of order 4.

(ii) Let $\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 4 & 8 & 6 & 7 & 5 & 1 & 3 \end{bmatrix}$ and

$$\beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 7 & 4 & 5 & 1 & 8 & 3 & 2 & 6 \end{bmatrix}$$

Write α , β and $\alpha\beta$ as product of 2-cycles.
(3+3=6)

(c) (i) Let $|a| = 24$. How many left cosets of $H = \langle a^4 \rangle$ in $G = \langle a \rangle$ are there? Write each of them.

(ii) State Fermat's Little theorem. Also compute $5^{25} \bmod 7$ and $11^{17} \bmod 7$.
(3+3=6)

4. (a) (i) Let H and K be two subgroups of a finite group. Prove that

$HK \leq G$ if G is Abelian.

(ii) Give an example of a group G and its two subgroups H and K ($H \neq K$) such that HK is not a subgroup of G .
(3+3.5=6.5)

(b) (i) Let G be a group and let $Z(G)$ be the centre of G . If $G/Z(G)$ is cyclic, prove that G is Abelian.

(ii) Let $|G| = pq$, p and q are primes. Prove that $|Z(G)| = 1$ or pq .
(4+2.5=6.5)

(c) (i) Prove that a subgroup of index 2 is normal.

(ii) Let $G = U(32)$, $H = U_8(32)$. Write all the elements of the factor group G/H . Also find order of $3H$ in G/H .
(3+3.5=6.5)

5. (a) Show that the mapping from $\mathbb{R}$ under addition to

$GL(2, \mathbb{R})$ that takes x to $\begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$ is a

group homomorphism. Also, find the kernel of the homomorphism.

(b) Let ϕ be a homomorphism from a group G to a group $\bar{G}$. Show that if $\bar{K}$ is a subgroup of $\bar{G}$,

then $\phi^{-1}(\bar{K}) = \{k \in G: \phi(k) \in \bar{K}\}$ is a subgroup of G .

(c) If H and K are two normal subgroups of a group G such that $H \subseteq K$, then prove that

$$G/K \approx \frac{G/H}{K/H} \quad (2 \times 6 = 12)$$

6. (a) Show that the mapping ϕ from $\mathbb{C}^*$ to $\mathbb{C}^*$ given by $\phi(z) = z^4$ is a homomorphism. Also find the set of all the elements that are mapped to 2.

(b) Prove that every group is isomorphic to a group of permutations.

(c) Let G be the group of non-zero complex numbers under multiplication and N be the set of complex numbers of absolute value 1.

Show that G/N is isomorphic to the group of all the positive real numbers under multiplication.

$$(2 \times 6.5 = 13)$$